

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Data Engineering and Software Engineering professional with **5+ years of experience** designing, developing, testing, and supporting scalable data solutions that translate complex business problems into reliable technical implementations.
- Advanced proficiency in **SQL, Python, Apache Spark, PySpark, and AWS**, with hands-on experience querying, transforming, integrating, and analyzing high-volume datasets containing hundreds of thousands to millions of records.
- Strong experience working as a technical partner to **business stakeholders, product teams, engineers, and analysts**, gathering requirements, asking investigative questions, defining solution approaches, and delivering data products aligned with business objectives.
- Experienced in **Big Data integration, ETL/ELT, distributed data processing, data modeling, API integrations, and cloud-based data pipelines**, consolidating information from heterogeneous sources into scalable analytical platforms.
- Proven expertise in **solution design, coding, quality testing, data validation, root-cause analysis, and production support**, ensuring implemented solutions function correctly against both technical requirements and real-world business needs.
- Experienced in preparing **technical specifications, solution documentation, data mappings, implementation plans, and code reviews**, while mentoring team members and maintaining quality and completeness across deliverables.
- Effective communicator and collaborative problem solver with experience working independently in **Agile/Scrum environments**, participating in daily standups, sprint planning, code reviews, frequent deployments, and cross-functional solution discussions.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Query Languages:** Python, SQL, PySpark, Java, Scala, C#, Bash, T-SQL, JavaScript, TypeScript.
- **Big Data & Processing:** Apache Spark, PySpark, Spark SQL, Hadoop, Distributed Data Processing, Batch Processing, Large-Scale Data Analysis, Data Integration, High-Volume Datasets.
- **Cloud Technologies:** Amazon Web Services (AWS), S3, Lambda, Glue, Redshift, EMR, EC2, CloudWatch, Azure, Docker, Kubernetes.
- **Data Engineering:** ETL/ELT, Data Pipelines, Data Modeling, Data Transformation, Data Cleansing, Data Validation, Data Reconciliation, Data Quality, Data Warehousing, API Integration.
- **Databases:** PostgreSQL, MySQL, Oracle, SQL Server, Amazon Redshift, Snowflake, SQL Query Optimization, Stored Procedures, CTEs, Window Functions.
- **Engineering & Quality:** Solution Design, Technical Specifications, Code Reviews, Unit Testing, Integration Testing, Quality Testing, Root-Cause Analysis, Performance Optimization, Production Support.
- **DevOps & Collaboration:** Git, GitHub, GitLab, Jenkins, TeamCity, Maven, Jira, Confluence, Splunk, CI/CD, Agile/Scrum, Sprint Planning, Daily Standups, Technical Documentation.

EXPERIENCE

Software Engineer

Oct 2024 – Present

PCMI LLC (Full Time)

Dallas, TX, USA

- Partnered with business, product, and engineering stakeholders to understand operational challenges, ask clarifying questions, define technical requirements, and **design data-driven solutions** that translated business objectives into scalable production implementations.
- Developed and optimized complex **SQL and Python solutions** to query, transform, reconcile, and analyze datasets containing hundreds of thousands to millions of contract, customer, claims, and transactional records across enterprise platforms.
- Built distributed data-processing workflows using **Apache Spark, PySpark, and Spark SQL**, implementing joins, aggregations, filtering, partitioning, and transformation logic to efficiently process high-volume datasets.
- Designed and maintained **ETL/ELT pipelines** integrating data from relational databases, REST APIs, flat files, and cloud storage into AWS-based analytical environments using S3, Glue, Lambda, and Redshift.
- Performed comprehensive **quality testing and data validation** by reconciling source and target records, validating business rules, identifying anomalies, testing edge cases, and working directly with stakeholders to confirm solutions worked as expected rather than only as designed.
- Investigated production data issues using SQL, Python, Splunk, and application logs, performing **root-cause analysis** and implementing high-quality fixes for data inconsistencies, pipeline failures, integration problems, and performance bottlenecks.
- Prepared **technical specifications, data mappings, solution designs, implementation documentation, and code-review feedback**, ensuring solutions were complete, maintainable, scalable, and aligned with defined business outcomes.
- Mentored engineers through technical discussions and solution reviews, collaborated within **Agile/Scrum sprints**, participated in daily standups and planning sessions, and supported frequent CI/CD deployments through TeamCity and Git-based workflows.

Environments: Python, SQL, Apache Spark, PySpark, Spark SQL, AWS S3, AWS Glue, AWS Lambda, Amazon Redshift, PostgreSQL, ETL/ELT, REST APIs, Data Integration, Data Modeling, Splunk, Git, TeamCity, Jira, Confluence, Agile/Scrum.

Software Engineer

Dec 2023 – Sept 2024

Client: CVS

Remote, USA

- Worked side-by-side with healthcare business stakeholders, analysts, product owners, and technical teams to understand business challenges and translate requirements into **functional data-engineering solutions** supporting customer and operational applications.
- Developed advanced **SQL and Python scripts** to extract, analyze, transform, and reconcile high-volume member, claims, pharmacy, provider, transaction, and application datasets across relational and cloud-based platforms.
- Leveraged **Apache Spark, PySpark, and Spark SQL** to process large datasets efficiently, applying distributed transformations, aggregations, joins, partitioning, and performance optimization for analytical and operational workloads.
- Designed cloud-based data pipelines using **AWS Lambda, S3, Glue, Redshift, REST APIs, and relational databases**, integrating information from multiple source systems into reusable downstream datasets.
- Implemented automated **data-quality and solution testing** covering completeness, accuracy, duplication, schema validation, referential integrity, business-rule validation, and source-to-target reconciliation prior to production deployment.
- Analyzed technical and business issues through SQL queries, Python utilities, API responses, and Splunk logs, identifying root causes and developing solutions that balanced **business objectives, data quality, performance, and maintainability**.
- Collaborated with engineering leads during **design reviews, code reviews, technical-specification reviews, and solution walkthroughs**, providing recommendations to improve completeness, scalability, reliability, and operational supportability.
- Delivered work through **Agile sprints**, participating in daily standups, sprint planning, refinement, retrospectives, CI/CD integration, and frequent deployment cycles while communicating progress and solution outcomes to technical and non-technical teams.

Environments: Python, SQL, Apache Spark, PySpark, AWS, AWS Lambda, S3, Glue, Amazon Redshift, MySQL, REST APIs, ETL/ELT, Data Integration, Data Quality, Splunk, GitLab, Jenkins, Jira, Confluence, Agile/Scrum.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Collaborated with product managers, business analysts, and engineering teams to understand enterprise application challenges and translate functional requirements into **data-processing, integration, and software solutions**.
- Developed complex **SQL queries, stored procedures, CTEs, joins, and aggregation logic** to retrieve and process high-volume transactional and application datasets supporting operational and analytical use cases.
- Built Python and **PySpark data-processing components** to cleanse, transform, aggregate, and integrate structured datasets, supporting scalable processing beyond traditional database workloads.
- Supported integration of data from **multiple relational databases, APIs, files, and enterprise applications** into centralized data-processing environments through ETL workflows and reusable transformation logic.
- Performed detailed **quality testing and backend data validation**, comparing expected business outcomes against actual data results and identifying discrepancies involving mappings, calculations, transformations, and application behavior.
- Investigated defects and production issues through SQL analysis, application logs, API debugging, and data reconciliation, performing **root-cause analysis** and collaborating with engineers until issues were resolved.
- Participated in technical design reviews and peer code reviews, validating **solution quality, completeness, maintainability, and alignment with business requirements** before deployment into shared environments.
- Worked within **two-week Agile sprints and frequent release cycles**, actively participating in daily standups, sprint planning, backlog refinement, defect triage, technical discussions, and deployment-readiness activities.

Environments: Python, SQL, PySpark, Apache Spark, Java, PostgreSQL, Oracle, MySQL, ETL, REST APIs, Data Integration, Data Validation, Jenkins, Git, Jira, Confluence, Splunk, Agile/Scrum.

Associate Software Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Worked with technical leads, analysts, and business stakeholders to understand application requirements and convert functional specifications into reliable **software and data-processing solutions**.
- Developed and maintained **SQL queries** to retrieve, validate, reconcile, and analyze transactional and application data from relational databases, supporting backend processing and business-rule verification.
- Used **Python and Java** to develop reusable data-processing utilities and application components for transforming records, validating datasets, and supporting system integration requirements.
- Supported **data integration from multiple source systems** including databases, APIs, and application interfaces, validating data movement and ensuring downstream systems received accurate and complete information.
- Performed functional, integration, system, regression, and **quality testing** to ensure implemented solutions satisfied technical specifications and expected business outcomes across application releases.
- Investigated application and data defects by analyzing SQL results, API responses, logs, and integration flows, documenting findings and collaborating with senior engineers to implement **root-cause fixes**.
- Participated in peer reviews, requirement discussions, and technical walkthroughs, incorporating feedback from senior engineers and contributing recommendations for improving **solution quality and completeness**.
- Worked within **Agile/Scrum teams**, participating in daily standups, sprint planning, story execution, defect tracking, CI/CD activities, and frequent release deadlines while communicating progress across distributed teams.

Environments: Python, SQL, Java, MySQL, Oracle, REST APIs, ETL, Data Integration, Data Validation, System Integration Testing, Git, Jenkins, Jira, Confluence, CI/CD, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.